CLOSED IMMERSIONS OF AN AFFINE SCHEME COME FROM IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1829-1847, following the statement at 1816-1827.

Editorial note. The closing sentence 'Of course I subset R and S = R/I and everything is clear' is the mechanical identification of the tilde modules with their global sections; the chain of engine citations before it is explicit.

Allowed standard prerequisites (editorial).

- Closed immersions as closed subspace structure. lemma-closed-immersion (same chapter, cited at ~1830). Identifies Z -> X with the closed subspace of a sheaf of ideals I.
- Quasi-coherence of the structure sheaf quotient. lemma-quasi-coherent-affine (same chapter, card L26380; section results consumed at ~1841-1842). O_X/I is of the form tilde(S) for an R-module S.
- Two out of three for quasi-coherence. lemma-extension-quasi-coherent (same chapter, card L26381, cited at ~1844). Surjectivity of tilde(R) -> tilde(S) forces I quasi-coherent.
- Model example of a closed immersion of affines. example-closed-immersion-affines (same chapter, cited at ~1826). The target identification Spec(R/I) -> Spec(R).

Source reference. Original definition 01HN: sources/stacks-schemes/pinned-original-source.tex, source lines 606–606. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01IG: sources/stacks-schemes/pinned-original-source.tex, source lines 1794–1794. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01HM: sources/stacks-schemes/pinned-original-source.tex, source lines 586–586. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01HO: sources/stacks-schemes/pinned-original-source.tex, source lines 617–617. Navigation pointer only: the counted proof cites

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this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01IE: sources/stacks-schemes/pinned-original-source.tex, source lines 1675–1675. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 01I6: sources/stacks-schemes/pinned-original-source.tex, source lines 1285–1285. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(X, \mathcal{O}_X) = (\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ be an affine scheme. Let $i : Z \to X$ be any closed immersion of locally ringed spaces. Then there exists a unique ideal $I \subset R$ such that the morphism $i : Z \to X$ can be identified with the closed immersion $\mathrm{Spec}(R/I) \to \mathrm{Spec}(R)$ constructed in Example 01IG above.*

Proof. This is kind of silly! Namely, by Lemma 01HO we can identify $Z \to X$ with the closed subspace associated to a sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$ as in Definition 01HN and Example 01HM. By our conventions this sheaf of ideals is locally generated by sections as a sheaf of $\mathcal{O}_X$-modules. Hence the quotient sheaf $\mathcal{O}_X/\mathcal{I}$ is locally on X the cokernel of a map $\bigoplus_{j \in J} \mathcal{O}_U \to \mathcal{O}_U$. Thus by definition, $\mathcal{O}_X/\mathcal{I}$ is quasi-coherent. By our results in Section 01I6 it is of the form $\widetilde{S}$ for some R-module S. Moreover, since $\mathcal{O}_X = \widetilde{R} \to \widetilde{S}$ is surjective we see by Lemma 01IE that also $\mathcal{I}$ is quasi-coherent, say $\mathcal{I} = \widetilde{I}$. Of course $I \subset R$ and $S = R/I$ and everything is clear. $\square$